Meaning of Dispersion

Dispersion refers to the variations of the items around an average. If the difference between the value of items and the average is large, the dispersion will be high and on the other hand if the difference between the value of items and the average is small, the dispersion will be low. Thus dispersion is defined as scatterness or spreadness of the individual items in a given series.

What is Measure of Dispersion

Measures of dispersion help to describe the variability in data. Dispersion is a statistical term that can be used to describe the extent to which data is scattered. Thus, measures of dispersion are certain types of measures that are used to quantify the dispersion of data.

Types of Measures of Dispersion

The measures of dispersion can be classified into two broad categories.

Absolute Measures of Dispersion: Absolute Measures of Dispersion is expressed in the same unit in which data of the series are expressed. They are expressed in the same statistical unit. Ex. Kilogram, Rupees, Tons, Meters, Years etc.

The most commonly used absolute measures of deviation are listed below.

1. Range
2. Quartile Deviation
3. Mean Deviation
4. Standard Deviation
5. Variance

Relative Measures of Dispersion: If the data of separate data sets have different units and need to be compared, then relative measures of dispersion are used. The measures are expressed in the form of ratios and percentages thus, making them unit less.

Some of the relative measures of dispersion are given below:

1. Coefficient of range
2. Coefficient of variation
3. Coefficient of standard deviation
4. Coefficient of quartile deviations

5. Coefficient of mean deviation

Range

It is a relative measure of the range.

Range (R) is the difference between the largest (L) and the smallest value (S) in a distribution.

$Range\ (R) = L - S.$

Coefficient of Range: It is a relative measure of the range. It is used in the comparative study of the dispersion

$$Coefficient\ of\ \text{Range} = \frac{L-S}{L+S}.$$

In case of continuous series, Range is just the difference between the upper limit of the highest class and the lower limit of the lowest class.

Example: The amount spent (in rupees) by the group of 10 students in the school canteen is as follows: 110, 117, 129, 197, 190, 100, 100, 178, 255, 790. Find the range and the co-efficient of the range.

Solution: Largest Value is L=790 and Smallest Value is S=100

Range $R = L - S = 790 - 100 = 690$

$$Coefficient\ of\ \text{Range} = \frac{L-S}{L+S} = \frac{790-100}{790+100} = 0.78.$$

Example: Calculate Range for the following data

Amount of taxes (in Thousand)	10	11	12	13	14
Frequency	3	12	18	12	3

Solution: Largest Value is L=10 and Smallest Value is S=14

Range $R = L - S = 14 - 10 = 4$

$$Coefficient\ of\ \text{Range} = \frac{L-S}{L+S} = \frac{4}{24} = 0.17.$$

Example: Find the range and it's co-efficient from the following data.

Size	10-20	20-30	30-40	40-50	50-100
Frequency	2	3	5	4	2

Solution: Range $R = L - S = 100 - 10 = 90$

$$Coefficient\ of\ \text{Range} = \frac{L-S}{L+S} = \frac{100-10}{100+10} = 0.82.$$

Example: Calculate the range for the following data: 60, 65, 70, 12, 52, 40, 48

Example: Calculate range for the following frequency distribution

Size	0-10	10-20	20-30	30-40	40-50
Frequency	2	6	12	7	3

Quartile Deviation

Quartile deviation can be defined as half of the difference between the third quartile and the first quartile in a given data set.

$$\text{Quartile Deviation (Q. D)} = \frac{Q_3 - Q_1}{2}$$

Coefficient of Quartile Deviation

$$\text{Coefficient of Quartile Deviation} = \frac{Q_3 - Q_1}{Q_3 + Q_1}.$$

In Individual Series: If only numbers are given, n is total number of observations.

$$\text{Q}_1 = \text{Value of } \left(\frac{\text{n}+1}{4}\right)^{\text{th}} \text{ term and } \text{Q}_3 = \text{Value of } \left(\frac{3(\text{n}+1)}{4}\right)^{\text{th}} \text{ term.}$$

Discrete Series: Data is given in the form of (x, f) and $N = \sum f$ is sum of all frequencies.

$$\text{Q}_1 = \text{Value of } \left(\frac{\text{N}+1}{4}\right)^{\text{th}} \text{ term and } \text{Q}_3 = \text{Value of } \left(\frac{3(\text{N}+1)}{4}\right)^{\text{th}} \text{ term.}$$

In Continues Distribution: Data is given in the form of class intervals and frequency:

$$Q_1 = l + h\frac{\left(\frac{N}{4} - C.F.\right)}{f} \text{ and } Q_3 = l + h\frac{\left(\frac{3N}{4} - C.F.\right)}{f}.$$

Where, l is lower limit of class interval

h is class width of class interval

$C.F.$ is cumulative frequency

f is frequency of class interval

$N = \sum f$ is sum of all frequencies

Example: Find out the quartile deviation and the coefficient of quartile deviation of daily wages (in ₹) of 7 persons is given below: 120, 70, 150, 100, 190, 170, 250.

Solution: Arranging the data in an ascending order we get 70, 100, 120, 150, 170, 190, 250

Here n = 7,

$Q_1 =$ Value of $\left(\frac{n+1}{4}\right)^{th}$ term $= \left(\frac{7+1}{4}\right)^{th} term = 2^{nd} term = 100\ rupees.$

$Q_3 =$ Value of $\left(\frac{3(n+1)}{4}\right)^{th}$ term $= \left(\frac{3(7+1)}{4}\right)^{th} term = 6^{th} term = 190\ rupees.$

Quartile Deviation $(Q.D) = \frac{Q_3 - Q_1}{2} = \frac{190-100}{2} = 45\ rupees.$

Coefficient of Quartile Deviation $= \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{190-100}{190+100} = \frac{90}{290} = \frac{9}{29}.$

Example: The tax Authority collected the following amount of tax from different firms in a particular market. Calculate the quartile deviation and the coefficient of quartile deviation

Amount of taxes (in Thousand)	10	11	12	13	14
Frequency	3	12	18	12	3

Solution:

Amount of taxes (x)	Frequency	Cumulative Frequency
10	3	3
11	12	15
12	18	33
13	12	45
14	3	48
	N=48	

$Q_1 =$ Value of $\left(\frac{N+1}{4}\right)^{th}$ term of x $= \left(\frac{48+1}{4}\right)^{th}$ term $= 12.25^{th}$ term $= 11$

$Q_3 =$ Value of $\left(\frac{3(N+1)}{4}\right)^{th}$ term $of\ x = \left(\frac{3(48+1)}{4}\right)^{th}$ term $= 36.75^{th}$ term $= 13$

Quartile Deviation $(Q.D) = \frac{Q_3 - Q_1}{2} = \frac{13-11}{2} = 1.$

Coefficient of Quartile Deviation $= \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{13-11}{13+11} = 0.083.$

Example: Find out the quartile deviation for the data

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency	3	5	7	9	4

Solution:

Class Interval	Frequency	Cumulative Frequency
0-10	3	3
10-20	5	8
20-30	7	15
30-40	9	24
40-50	4	28
	N=28	

For Q_1: Find $\frac{N}{4} = \frac{28}{4} = 7$ and $l, h, f, C.F.$ from Class Interval (c.f. greater than or equal to 7 is 8) hence Class Interval is 10-20 and $l = 10, f = 5, h = 10, C.F. = 3,$

$$Q_1 = l + h\frac{\left(\frac{N}{4} - C.F.\right)}{f} = 10 + 10\left(\frac{7-3}{5}\right) = 18$$

For Q_3: $\frac{3N}{4} = 21$, (therefore c.f. greater than or equal to 21 is 24) hence Class Interval is 30-40 and $l = 30, f = 9, h = 10, C.F. = 15,$

$$Q_3 = l + h\frac{\left(\frac{3N}{4} - C.F.\right)}{f} = 30 + 10\left(\frac{21-15}{9}\right) = 36.67$$

$$\text{Quartile Deviation (Q. D)} = \frac{Q_3 - Q_1}{2} = \frac{36.67-18}{2} = 9.335.$$

Mean Deviation

Mean deviation (MD) of a series is the arithmetic average of the deviation of various items from a measure of central points (mean, median and mode).

Type	Mean deviation about Assumed Mean	Mean deviation about Median	Mean deviation about Arithmetic Mean
Individual Series	M.D.= $\frac{1}{n}\sum_{i=0}^{n}\lvert x_i - A\rvert$ Where A is Assumed mean	M.D.= $\frac{1}{n}\sum_{i=0}^{n}\lvert x_i - Median\rvert$	M.D.= $\frac{1}{n}\sum_{i=0}^{n}\lvert x_i - \bar{x}\rvert$ where $\bar{x}$ is arithmetic mean

Discrete & Continuous Series	M.D.=$\frac{1}{N}\sum_{i=0}^{n} f\lvert x_i - A\rvert$ Where $N=\sum f$ & A is Assumed mean	M.D.=$\frac{1}{N}\sum_{i=0}^{n} f\lvert x_i - Median\rvert$ Where $N=\sum f$	M.D.=$\frac{1}{N}\sum_{i=0}^{n} f\lvert x_i - \bar{x}\rvert$ where $\bar{x}$ is arithmetic mean and $N=\sum f$

Coefficient of Mean Deviation: This can be defined as the ratio of the mean deviation to the value of the central point from which it is calculated.

$$\text{Coefficient of Mean Deviation} = \frac{M.D.}{Mean/Median/Mode}$$

Example: Find Mean Deviation of 120, 70, 150, 100, 190, 170, 250.

Solution: First find $\bar{x} = \frac{120+70+150+100+190+170+250}{7} = \frac{1050}{7} = 150$

x	$\bar{x}$	$x-\bar{x}$	$\lvert x-\bar{x}\rvert$
70	150	-80	80
100	150	-50	50
120	150	-30	30
150	150	0	0
170	150	20	20
190	150	40	40
250	150	100	100
			320

M.D.=$\frac{1}{n}\sum_{i=0}^{n}\lvert x_i - \bar{x}\rvert = \frac{320}{7} = 45.7.$

Example: Calculate Mean Deviation from mean for the following data

x	1	2	3	4	5	6	7
f	3	5	8	12	10	7	5

Solution:

x	f	fx	$\lvert x-\bar{x}\rvert$	$f\lvert x-\bar{x}\rvert$
1	3	3	3.24	9.72
2	5	10	2.24	11.2
3	8	24	1.24	9.92

4	12	48	0.24	2.88
5	10	50	0.76	7.6
6	7	42	1.76	12.32
7	5	35	2.76	13.8
	50	**212**		**67.44**

Mean $\bar{x} = \frac{\sum fx}{\sum f} = \frac{212}{50} = 4.24$

M.D.=$\frac{1}{N}\sum_{i=0}^{n} f|x_i - \bar{x}| = \frac{67.44}{50} = 1.348.$

Example: Calculate mean deviaiton and coefficient of mean deviation from both mean and median for the following data on the monthly income (In Rs) of households.

Monthly Income: 8520, 6350, 7920, 8360, 7500.

Solution: Given $n = 5$ numbers are in ascending order: 6350, 7500, 7920, 8360, 8520

$\text{Median} = \text{Value of } \left(\frac{n+1}{2}\right)^{th} term = 3^{rd}\ term = 7920.$

$\text{Mean} = \frac{6350+7500+7920+8360+8520}{5} = \frac{38650}{5} = 7730.$

Monthly Income	$\lvert x_i - \bar{x}\rvert$	$\lvert x_i - Median\rvert$
6350	1380	1570
7500	230	420
7920	190	0
8360	630	440
8520	790	600
	3220	3030

Mean deviation about Mean: M.D.=$\frac{1}{n}\sum_{i=0}^{n}|x_i - \bar{x}| = \frac{3220}{5} = 664.$

$\text{Coefficient of Mean Deviation} = \frac{M.D.}{Mean} = \frac{664}{7730} = 0.083.$

Mean deviation about Median: M.D.=$\frac{1}{\text{n}}\sum_{i=0}^{n}|x_i - Median| = \frac{3030}{5} = 606.$

$\text{Coefficient of Mean Deviation} = \frac{M.D.}{Median} = \frac{606}{7920} = 0.076.$

Example : Calculate (i) mean, (ii) mean deviation from mean and (iii) co-efficient of mean deviation.

Marks	0-10	10-20	20-30	30-40	40-50
No. of students	5	8	15	16	6

Solution: Calculation of mean deviation from mean

C.I.	**Mid- point x**	**f**	**fx**	$\|x- 27\|$	$f\|x- 27\|$
0-10	5	5	25	22	110
10-20	15	8	120	12	96
20-30	25	15	375	2	30
30-40	35	16	560	8	128
40-50	45	6	270	18	108
		N=50	1350		472

Mean $\bar{x} = \frac{1350}{50} = 27$

Mean deviation about Mean: M.D.=$\frac{1}{N}\sum_{i=0}^{n} f|x_i - \bar{x}| = \frac{472}{50} = 9.44.$

Coefficient of Mean Deviation $= \frac{M.D.}{Mean} = \frac{9.44}{472} = 0.349.$

Example: Find the Mean deviation from median for the following data

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency	3	5	7	9	4

Solution: Calculation of mean deviation from mean

C.I.	**Mid- point x**	**f**	**C.F.**	$f\|x- median\|$
0-10	5	3	3	70.71
10-20	15	5	8	67.85
20-30	25	7	15	24.99
30-40	35	9	24	57.87
40-50	45	4	28	65.72
		N=28		287.14

$\frac{N}{2} = 14$, The 14th observation falls in the class interval 20-30.

Hence $l = 20, h = 10, f = 7, C.F. = 8$

$$\text{Median} = l + h\frac{\left(\frac{N}{2} - C.F.\right)}{f} = 20 + 10\frac{14-8}{7} = 28.57$$

$$\text{M.D.} = \frac{1}{N}\sum_{i=0}^{n} f|x_i - Median| = \frac{287.14}{28} = 10.255.$$

Example: Calculated mean deviation and co-efficient of mean deviation from median

No. of tomatoes per plant	0	1	2	3	4	5	6	7	8	9	10
No. of plants	2	5	7	11	18	24	12	8	6	4	3

Example: Calculate mean deviation from mean

Class:	3-4	4-5	5-6	6-7	7-8	8-9	9-10
Frequency	3	7	22	60	85	32	8

Standard Deviation

The square root of the variance gives the standard deviation. Thus, the standard deviation also measures the variation of the data about the mean.

1. A. Standard Deviation without frequency: $\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2} = \sqrt{\frac{1}{n}\sum x^2 - (\overline{x})^2}$

   B. Standard Deviation with frequency: $\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{n} f(x_i - \overline{x})^2} = \sqrt{\frac{1}{N}\sum f(x - \overline{x})^2}$

2. If $d = x - A$ then $\sigma = \sqrt{\frac{\sum f \cdot d^2}{N} - \left(\frac{\sum f \cdot d}{N}\right)^2}$

3. If $u = \frac{x-A}{h}$ where 'h' is width of class, then $\sigma = h\sqrt{\frac{\sum f \cdot u^2}{N} - \left(\frac{\sum f \cdot u}{N}\right)^2}$

Variance: The average of the squared differences from the mean or $v = \sigma^2$.

A. Variance without frequency: $v = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2 = \frac{1}{n}\sum x^2 - (\overline{x})^2$

B. Variance with frequency: $v = \frac{1}{N}\sum_{i=1}^{n} f(x_i - \overline{x})^2 = \frac{1}{N}\sum f(x - \overline{x})^2$

Coefficient of Variation: $C.V = \frac{S.D}{|A.M|} \times 100 = \frac{\sigma}{\overline{x}} \times 100$

Comparison of two objects A and B: If $(\mathrm{C.V})_{\mathrm{A}} < (\mathrm{C.V})_{\mathrm{B}}$ then

- Object A is more consistent. Or Object B is more variable.

Variance

The average squared deviation from the mean of the given data set is known as the variance. This measure of dispersion checks the spread of the data about the mean.

Coefficient of Variation: It is the ratio of the standard deviation to the mean of the data set. It is expressed in the form of a percentage.

Example: Runs scored in 10 matches of IPL season by two batsmen A and B are tabulated as under

Batsman A	46	34	52	78	65	81	26	46	19	47
Batsman B	59	25	81	47	73	78	42	35	42	10

Decide who is better batsman and who is more consistent.

Solution: Find standard deviation for both batsmen

											Total
Batsman A	x	46	34	52	78	65	81	26	46	19	47	**494**
	x^2	2116	1156	2704	6084	4225	6561	676	2116	361	2209	**28208**

Batsman B	x	59	25	81	47	73	78	42	35	42	10	**492**
	x^2	3481	625	6561	2209	5329	6084	1764	1225	1764	100	**29142**

For Batsman A: $\overline{x}_A = \frac{\sum x}{n} = 49.4$

For Batsman B: $\overline{x}_B = \frac{\sum x}{n} = 49.2$

For Batsman A: $\sigma_A = \sqrt{\frac{1}{n}\sum x^2 - (\overline{x})^2} = \sqrt{\frac{28208}{10} - (49.4)^2} = 19.50$

For Batsman B: $\sigma_B = \sqrt{\frac{1}{n}\sum x^2 - (\overline{x})^2} = \sqrt{\frac{29142}{10} - (49.2)^2} = 22.22$

$(C.V)_A = \frac{S.D}{|A.M|} \times 100 = \frac{\sigma_A}{\overline{x}} \times 100 = 39.47$

$(C.V)_B = \frac{S.D}{|A.M|} \times 100 = \frac{\sigma_B}{\overline{x}} \times 100 = 45.16$

$(\mathrm{C.V})_{\mathrm{A}} < (\mathrm{C.V})_{\mathrm{B}}$

Conclusion: A is more consistent.

Example: Calculate standard deviation for the following frequency distribution. Decide whether A.M. is good average.

Wages in Rupees earned per day	0-10	10-20	20-30	30-40	40-50	50-60
No. of laborers	5	9	15	12	10	3

Solution: Preparing the table for the purpose of calculations. Assume A=25

C.I.	Mid Values (x)	f	$u = \frac{x - A}{h}$	fu	fu^2
0-10	5	5	-2	-10	20
	15	9	-1	-9	9
20-30	25	15	0	0	0
30-40	35	12	1	12	12
40-50	45	10	2	20	40
50-60	55	3	3	9	27
		54		22	108

$$\sigma = h\sqrt{\frac{\sum f \cdot u^2}{N} - \left(\frac{\sum f \cdot u}{N}\right)^2} = 10\sqrt{\frac{108}{54} - \left(\frac{22}{54}\right)^2} = 13.54$$

$$A.M. = A + h\frac{\sum fu}{\sum f} = 25 + 10\frac{22}{54} = 29.07$$

Conclusion: $\sigma = 13.54$ is quite a large value and Arithmetic mean 29.074 is not a good average.

Example: Fluctuations in the Aggregate of marks obtained by two groups of students are given below. Find out which of the two shows greater variability.

Group A	518	519	530	530	544	542	518	550	527	527	531	550	550	529	528
Group B	825	830	830	819	814	814	814	844	842	842	826	832	835	840	840

Solution: To solve this problem, we have to determine coefficient of variation each case. First we present the data in frequency distribution form.

Group A	
518	268324
519	269361
530	280900
530	280900
544	295936

Group B	
825	680625
830	688900
830	688900
819	670761
814	662596

542	293764		814	662596
518	268324		814	662596
550	302500		844	712336
527	277729		842	708964
527	277729		842	708964
531	281961		826	682276
550	302500		832	692224
550	302500		835	697225
529	279841		840	705600
528	278784		840	705600
7993	**4261053**		**12447**	**10330163**

For Group A: $\overline{x}_A = \frac{\sum x}{n} = 532.87$

For GroupB: $\overline{x}_B = \frac{\sum x}{n} = 829.80$

For Batsman A: $\sigma_A = \sqrt{\frac{1}{n}\sum x^2 - (\overline{x})^2} = 11.10$

For Batsman B: $\sigma_B = \sqrt{\frac{1}{n}\sum x^2 - (\overline{x})^2} = 10.46$

$(C.V)_A = \frac{S.D}{|A.M|} \times 100 = 2.08$

$(C.V)_B = \frac{S.D}{|A.M|} \times 100 = 1.158$

$(\mathrm{C.V})_\mathrm{A} > (\mathrm{C.V})_\mathrm{B}$

Conclusion: Group A has greater variability, or Group B is more consistent.

Example: Arithmetic mean and standard deviation of 30 items are 20 and 3 respectively out of these 30 items, item 22 and 15 are dropped. Find new A.M. and S.D. Calculate A.M. and S.D. if item 22 is replaced by 8 and 15 is replaced by 17.

Solution: Given that $n = 30$, A.M. $\overline{x} = 20$ and $S.D. = 3$. $\overline{x} = \frac{\sum x}{n} \rightarrow \sum x = n\,\overline{x} = 30 \times 20 = 600$

Since item 22 and 15 are dropped, $\sum x = 600 - 22 - 15 = 563$ and total items are $n = 28$, hence $\overline{x} = \frac{\sum x}{n} = \frac{563}{28} = 20.107$.

Standard deviation $\sigma = \sqrt{\frac{\sum x^2}{n} - (\overline{x})^2} = \sqrt{\frac{\sum x^2}{30} - (20)^2} \rightarrow\rightarrow 3 = \sqrt{\frac{\sum x^2}{30} - 400}$

$\sum x^2 = 30(9 + 400) = 12270$

But item 22 and 15 are dropped $\sum x^2 = 12270 - (22)^2 - (15)^2 = 11561$

Again standard deviation $\sigma = \sqrt{\frac{\sum x^2}{n} - (\overline{x})^2} = \sqrt{\frac{11561}{28} - (20.107)^2} = 2.933$

In the second case if items are replaced

$\sum x = 600 - 22 + 8 - 15 + 17 = 588$ and $n = 30$

Now $\overline{x} = \frac{\sum x}{n} = \frac{588}{30} = 19.6$

$\sum x^2 = 12270 - (22)^2 + (8)^2 - (15)^2 + (17)^2 = 12078$

$\sigma = \sqrt{\frac{\sum x^2}{n} - (\overline{x})^2} = \sqrt{\frac{12078}{30} - (19.6)^2} = 4.294.$

Moments

A moment is a quantitative measurement for the shape of a function.

The shape of any distribution can be described by its various 'moments'.

The first four are:

1. The mean, which indicates the central tendency of a distribution.
2. The second moment is the variance, which indicates the width or deviation.
3. The third moment is the skewness, which indicates any asymmetric 'leaning' to either left or right.
4. The fourth moment is the Kurtosis, which indicates the degree of central 'Peaked ness' or, equivalently, the 'fatness' of the outer tails.

Central Moment

Moments about mean are called central moments.

- The rth moment about mean without frequency: $\mu_r = \frac{1}{n}\sum(x - \overline{x})^r$
- The rth moment about mean with frequency: $\mu_r = \frac{1}{N}\sum f(x - \overline{x})^r \ \ and \ N = \sum f$
  - If $r = 0$, then $\mu_0 = 1$.
  - If $r = 1$, then $\mu_1 = 0$.
  - If $r = 2$, then $\mu_2 = \sigma^2 = v$.

Raw Moments

Moment about any point of observations different from mean is called as raw moments.

- The rth moment about any value of A without frequency: $\mu'_r = \frac{1}{n}\sum(x-A)^r$
- The rth moment about any value of A with frequency: $\mu'_r = \frac{1}{N}\sum f(x-A)^r \; and \; N = \sum f$.

Remarks:

- If $r = 0$then $\mu'_0 = 1$.
- If $r = 1$ then $\mu'_1 = \overline{x} - A \Rightarrow Mean$ is $\overline{x} = \mu'_1 + A$.

Alternate Formula: $\mu'_r = h^r \frac{1}{N}\sum(f \cdot u^r) = h^r \frac{\sum(fu^r)}{N}$

Where A= assumed Mean, h-width of class, $u = \frac{x-A}{h}$ and r=1, 2, 3…

Relation Between μ_r and μ'_r:

- $\mu_1 = 0$
- $\mu_2 = \mu'_2 - (\mu'_1)^2$
- $\mu_3 = \mu'_3 - 3(\mu'_2)(\mu'_1) + 2(\mu'_1)^3$
- $\mu_4 = \mu'_4 - 4(\mu'_3)(\mu'_1) + 6(\mu'_2)(\mu'_1)^2 - 3(\mu'_1)^4$
- $General \; formula \; \mu_r = \mu'_r - rc_1(\mu'_{r-1})(\mu'_1) + rc_2(\mu'_{r-2})(\mu'_1)^2 - \cdots + (-1)^r(\mu'_1)^r$

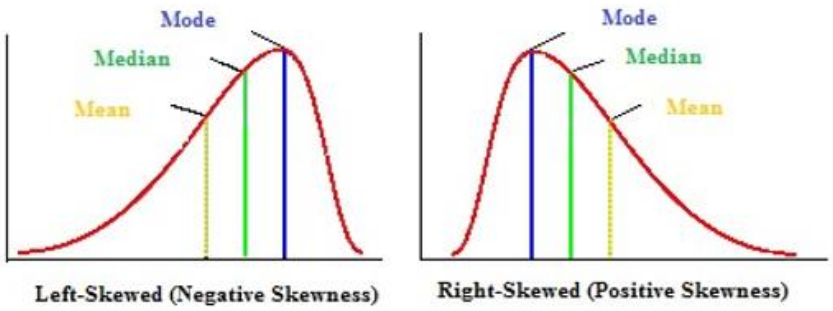
Skewness

The different measures of skewness are:

1. Skewness $= 3\frac{(Mean - Median)}{Standard \; Deviation}$
2. Coefficient of Skewness: $\beta_1 = \frac{\mu_3^2}{\mu_2^3} \; or \; \gamma_1 = +\sqrt{\beta_1}$

Positive Skewness: (Mode < Median <Mean) If the mean lies to the right side of mode, then the frequency curve stretches to the right then the distribution is right skewed or positively skewed.

Negative Skewness: (Mode > Median >Mean) If the mean lies to the left side of mode, then the frequency curve stretches to the left then the distribution is right skewed or positively skewed.



Kurtosis

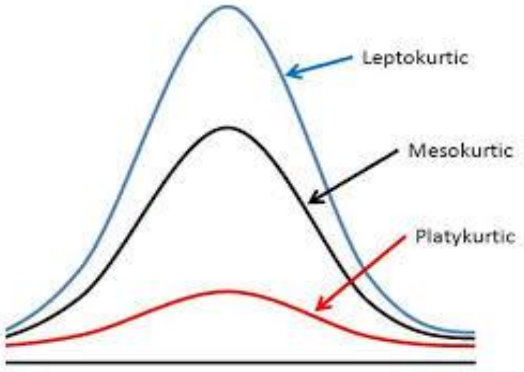
Kurtosis (Convexity of curve) is a measure which gives an idea about the flatness or peakedness of the curve. It is measured by the following coefficient.

Coefficient of Kurtosis: $\beta_2 = \frac{\mu_4}{\mu_2^2}$ or $\gamma_2 = \beta_2 - 3$

Leptokurtic Curve: The curve which is more peaked for which $\beta_2 > 3$

Mesokurtic (Normal) Curve: The curve which is neither flat nor peaked for which $\beta_2 = 3$

Platykurtic Curve: The curve which is flatter for which $\beta_2 < 3$.



Examples:

Ex1. The first three moments of a distribution about the value 2 are 1, 16 and -40. Find mean, standard deviation and $\beta_1$of the distribution.

Solution: $Let\ A = 2,\ \mu'_1 = 1, \mu'_2 = 16, \mu'_3 = -40\ and\ \mu_1 = 0$

$\mu_2 = \mu'_2 - (\mu'_1)^2 = 16 - (1)^2 = 15$

$\mu_3 = \mu'_3 - 3(\mu'_2)(\mu'_1) + 2(\mu'_1)^3 = -40 - 3(16)(1) + 2(1)^3 = -40 - 48 + 2 = -86$

$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(-86)^2}{(15)^3} = \frac{7396}{225} = 32.87$

Consider $\mu'_1 = \overline{x} - A \Rightarrow Mean$ is $\overline{x} = \mu'_1 + A = 1 + 2 = 3$ and use $\mu_2 = \sigma^2 \Rightarrow 15 = \sigma^2$

Standard deviation is $\sigma = \pm\sqrt{15}$.

Ex2. The first four moments of a distribution about the value 4 are -1.5, 17, -30 and 108. Obtain the first four central moments, mean, standard deviation and co-efficient of skewness and kurtosis.

Solution: Let $A = 4, \mu'_1 = -1.5, \mu'_2 = 17, \mu'_3 = -30, \mu'_4 = 108$ and $\mu_1 = 0$

$\mu_2 = \mu'_2 - (\mu'_1)^2 = 17 - (-1.5)^2 = 14.75$

$\mu_3 = \mu'_3 - 3(\mu'_2)(\mu'_1) + 2(\mu'_1)^3 = -30 - 3(17)(-1.5) + 2(-1.5)^3 = 39.75$

$\mu_4 = \mu'_4 - 4(\mu'_3)(\mu'_1) + 6(\mu'_2)(\mu'_1)^2 - 3(\mu'_1)^4 = 362.81$

$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(39.75)^2}{(14.75)^3} = 0.4923$ $\qquad$ $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{362.81}{(14.75)^2} = 1.667$

Consider $\mu'_1 = \overline{x} - A \Rightarrow Mean$ is $\overline{x} = \mu'_1 + A = -1.5 + 4 = 2.5$ and

use $\mu_2 = \sigma^2 \Rightarrow 14.75 = \sigma^2 \Rightarrow \sigma = \pm\sqrt{14.75}$.

Ex3. The first four moments of distribution about the value 5 are 3, 30, 50, and 60. Obtain the first four central moments and co-efficient of skewness and kurtosis.

Solution: Let $A = 5, \mu'_1 = 3, \mu'_2 = 30, \mu'_3 = 50, \mu'_4 = 60$ and $\mu_1 = 0$

$\mu_2 = \mu'_2 - (\mu'_1)^2 = 30 - (3)^2 = 21$

$\mu_3 = \mu'_3 - 3(\mu'_2)(\mu'_1) + 2(\mu'_1)^3 = 500 - 3(30)(3) + 2(3)^3 = 284$

$\mu_4 = \mu'_4 - 4(\mu'_3)(\mu'_1) + 6(\mu'_2)(\mu'_1)^2 - 3(\mu'_1)^4 = 1647$

$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(284)^2}{(21)^3} = 8.709$ $\qquad$ $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{1647}{(21)^2} = 3.73$

Consider $\mu'_1 = \overline{x} - A \Rightarrow Mean$ is $\overline{x} = \mu'_1 + A = 3 + 4 = 7$

And $\mu_2 = \sigma^2 \Rightarrow 21 = \sigma^2 \Rightarrow \sigma = \pm\sqrt{21}$.

Ex4. Find the first four moments of the following distribution about the mean and hence findβ_1, β_2.

Marks obtained	0-10	10-20	20-30	30-40	40-50	50-60
Number of students	1	6	10	15	11	7

Solution: Let $h = 10, A = 35$ (value from x column)

(C.I)	Frequency f	Mid values x	$u = \frac{x - A}{h}$	$f \cdot u$	$f \cdot u^2$	$f \cdot u^3$	$f \cdot u^4$

0-10	1	5	-3	-3	9	-27	81
10-20	6	15	-2	-12	24	-48	96
20-30	10	25	-1	-10	10	-10	10
30-40	15	35	0	0	0	0	0
40-50	11	45	1	11	11	11	11
50-60	7	55	2	14	28	56	112
Total	N=50			0	82	-18	310

Use formula $\mu'_r = \frac{h^r \sum f \cdot u^r}{N}$

$\mu'_1 = \frac{h^1 \sum f \cdot u^1}{N} = \frac{10*0}{50} = 0$ $\mu'_2 = \frac{h^2 \sum f \cdot u^2}{N} = \frac{(10)^2*82}{50} = 164$

$\mu'_3 = \frac{h^3 \sum f \cdot u^3}{N} = \frac{(10)^3*(-18)}{50} = -360$ $\mu'_4 = \frac{h^4 \sum f \cdot u^4}{N} = \frac{(10)^4*(310)}{50} = 62000$

Moments about mean are $\mu_1 = 0$

$\mu_2 = \mu'_2 - (\mu'_1)^2 = 160$

$\mu_3 = \mu'_3 - 3(\mu'_2)(\mu'_1) + 2(\mu'_1)^3 = -360$

$\mu_4 = \mu'_4 - 4(\mu'_3)(\mu'_1) + 6(\mu'_2)(\mu'_1)^2 - 3(\mu'_1)^4 = 62000$

$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0.03164$ $\beta_2 = \frac{\mu_4}{\mu_2^2} = 2.4218.$

Ex5. Calculate the first four moments about mean of the distribution. Also findβ_1, β_2

x	2.0	2.5	3.0	3.5	4.0	4.5	5.0
f	4	36	60	90	70	40	10

Solution: Let h=0.5, A=3.5 (value from x column)

x	f	$u = \frac{x - A}{h}$	fu	fu^2	fu^3	fu^4
2	4	-3	-12	36	-108	324
2.5	36	-2	-72	144	-288	576
3	60	-1	-60	60	-60	60
3.5	90	0	0	0	0	0
4	70	1	70	70	70	70
4.5	40	2	80	160	320	640

5	10	3	30	90	270	810
Total	N=310		36	560	204	2480

Use formula $\mu'_r = \frac{h^r \sum f \cdot u^r}{N}$

$\mu'_1 = \frac{h^1 \sum f \cdot u^1}{N} = \frac{0.5*36}{310} = 0.05806$ $\mu'_2 = \frac{h^2 \sum f \cdot u^2}{N} = \frac{(0.5)^2*560}{5310} = 0.4516$

$\mu'_3 = \frac{h^3 \sum f \cdot u^3}{N} = \frac{(0.5)^3*(204)}{310} = 0.0822$ $\mu'_4 = \frac{h^4 \sum f \cdot u^4}{N} = \frac{(0.5)^4*2480}{310} = 0.5$

Moments about mean are $\mu_1 = 0$

$\mu_2 = \mu'_2 - (\mu'_1)^2 = 0.4482$

$\mu_3 = \mu'_3 - 3(\mu'_2)(\mu'_1) + 2(\mu'_1)^3 = 0.003963$

$\mu_4 = \mu'_4 - 4(\mu'_3)(\mu'_1) + 6(\mu'_2)(\mu'_1)^2 - 3(\mu'_1)^4 = 0.4899$

$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0.000174$ $\beta_2 = \frac{\mu_4}{\mu_2^2} = 2.43878.$

Ex6. Calculate the first four central moments for the following distribution. Find the coefficient of Skewness and Kurtosis.

X	1	2	3	4	5	6	7	8	9	10
f	6	15	23	42	62	60	40	24	13	5

Solution: Let h=1, A=5 (value from x column)

x	f	$u = \frac{x - A}{h}$	fu	fu^2	fu^3	fu^4
1	6	-4	-24	96	-384	1536
2	15	-3	-45	135	-405	1215
3	23	-2	-46	92	-184	368
4	42	-1	-42	42	-42	42
5	62	0	0	0	0	0
6	60	1	60	60	60	60
7	40	2	80	160	320	640
8	24	3	72	216	648	1944
9	13	4	52	208	832	3328
10	5	5	25	125	625	3125

Total	N=290	5	132	1134	1470	12258

Use formula $\mu'_r = \frac{h^r \sum f \cdot u^r}{N}$

$\mu'_1 = \frac{h^1 \sum f \cdot u^1}{N} = \frac{1*132}{290} = 0.4551$ $\mu'_2 = \frac{h^2 \sum f \cdot u^2}{N} = \frac{(1)^2*1134}{290} = 3.9103$

$\mu'_3 = \frac{h^3 \sum f \cdot u^3}{N} = \frac{(1)^3*1470}{290} = 5.0689$ $\mu'_4 = \frac{h^4 \sum f \cdot u^4}{N} = \frac{(1)^4*12258}{290} = 42.2689$

Moments about mean are $\mu_1 = 0$

$\mu_2 = \mu'_2 - (\mu'_1)^2 = 0.4482$

$\mu_3 = \mu'_3 - 3(\mu'_2)(\mu'_1) + 2(\mu'_1)^3 = 0.003963$

$\mu_4 = \mu'_4 - 4(\mu'_3)(\mu'_1) + 6(\mu'_2)(\mu'_1)^2 - 3(\mu'_1)^4 = 0.49$

$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0.000174.$ $\beta_2 = \frac{\mu_4}{\mu_2^2} = 2.439.$